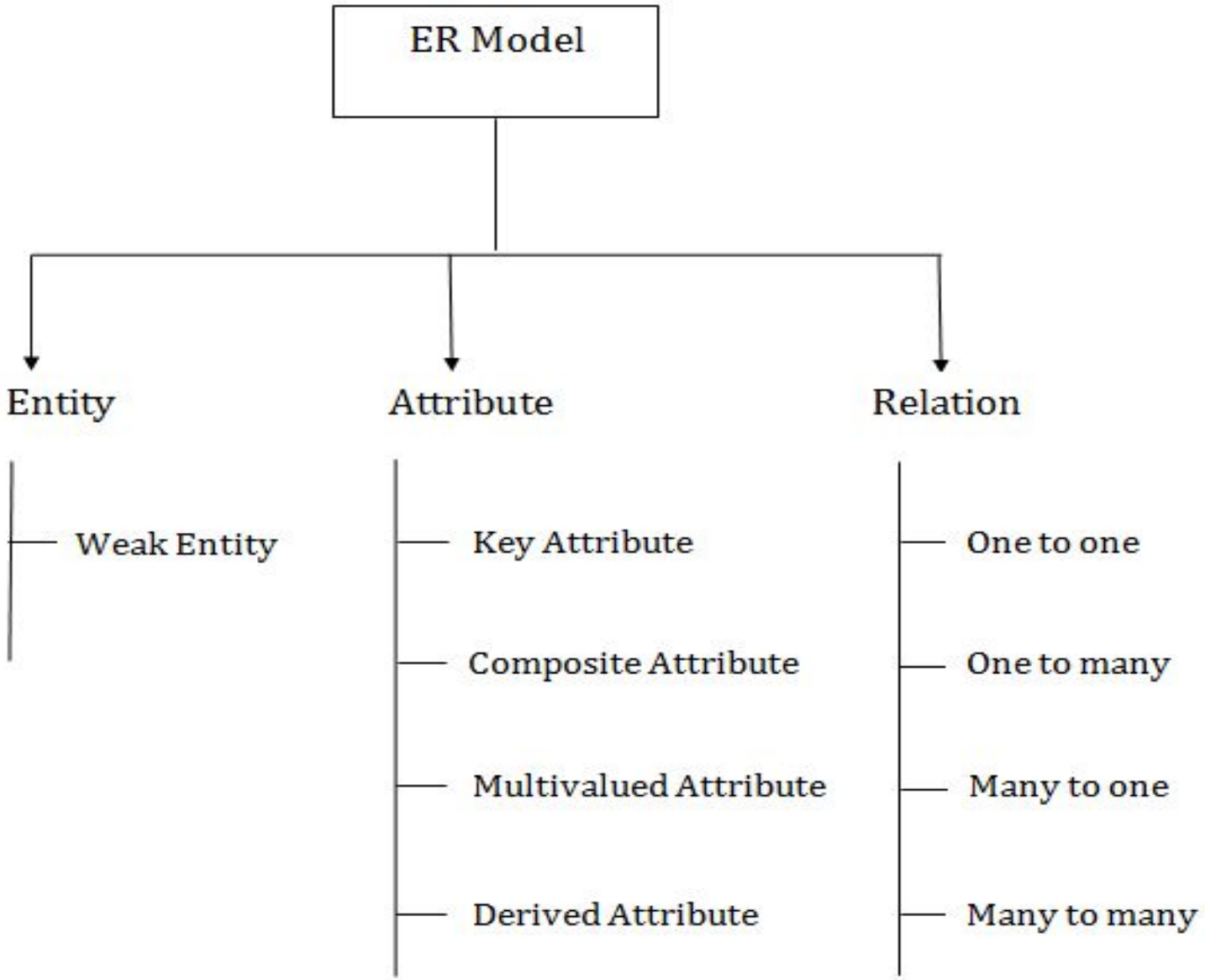
E-R Model
(Entity – Relationship)

Entity-Relationship model

- ER model stands for an Entity-Relationship model.
- It is a high-level data model.
- This model is used to define the data elements and relationship for a specified system.
- It develops a conceptual design for the database.
- It also develops a very simple and easy to design view of data.
- In ER modeling, the database structure is portrayed as a diagram called an entity-relationship diagram.

Components of E-R model



Entity:

- An entity may be any object, class, person or place.
- In the ER diagram, an entity can be represented as rectangles.

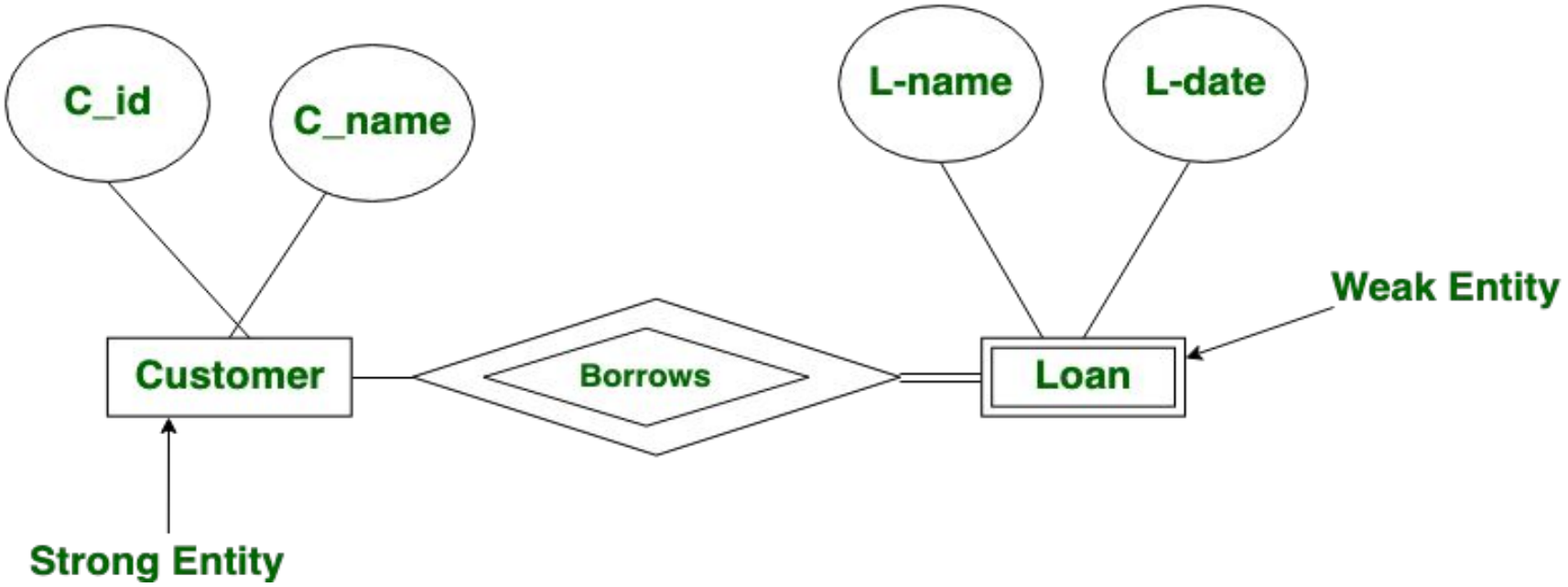
Attribute:

- The attribute is used to describe the property of an entity. Eclipse is used to represent an attribute.

Relationship:

- A relationship is used to describe the relation between entities. Diamond or rhombus is used to represent the relationship.

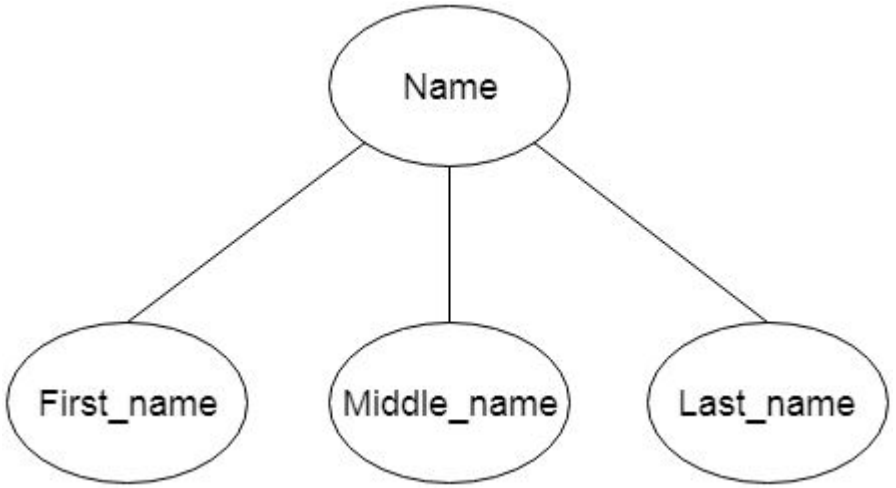
- **Strong Entity:**
- A strong entity is not dependent on any other entity in the schema.
- A strong entity will always have a primary key. Strong entities are represented by a single rectangle.
- The relationship of two strong entities is represented by a single diamond.
- Various strong entities, when combined together, create a strong entity set.
- **Weak Entity:**
- A weak entity is dependent on a strong entity to ensure its existence. Unlike a strong entity, a weak entity does not have any primary key.
- It instead has a partial discriminator key.
- A weak entity is represented by a double rectangle.
- The relation between one strong and one weak entity is represented by a double diamond.
- This relationship is also known as **identifying relationship.**



Types of attributes:

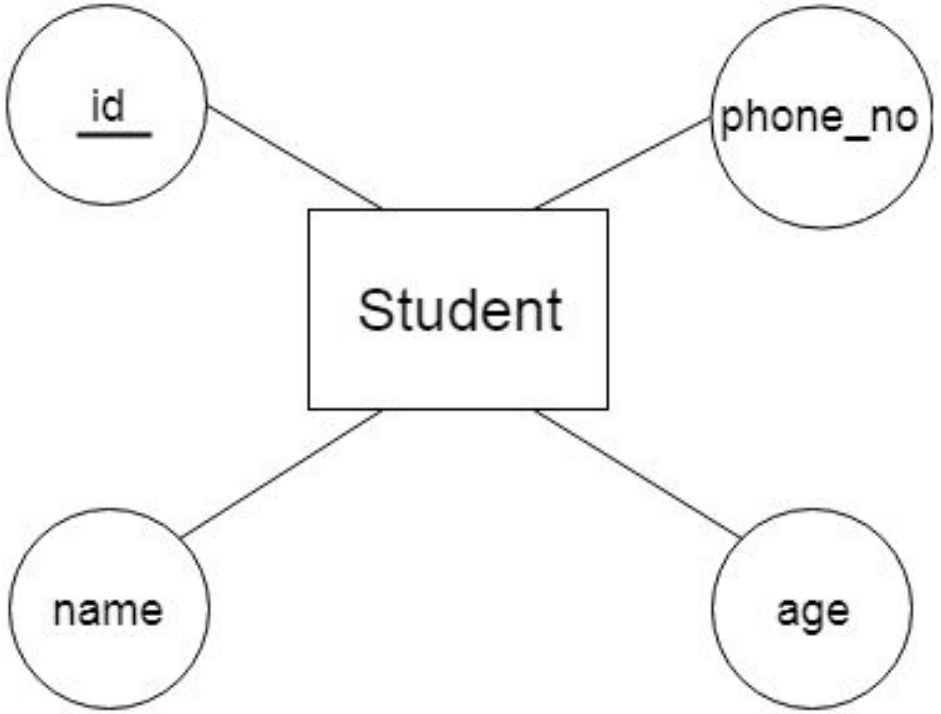
Key Attribute

The key attribute is used to represent the main characteristics of an entity. It represents a primary key. The key attribute is represented by an ellipse with the text underlined.



b. Composite Attribute

An attribute that composed of many other attributes is known as a composite attribute. The composite attribute is represented by an ellipse, and those ellipses are connected with an ellipse.



c. Multivalued Attribute

An attribute can have more than one value. These attributes are known as a multivalued attribute. The double oval is used to represent multivalued attribute.

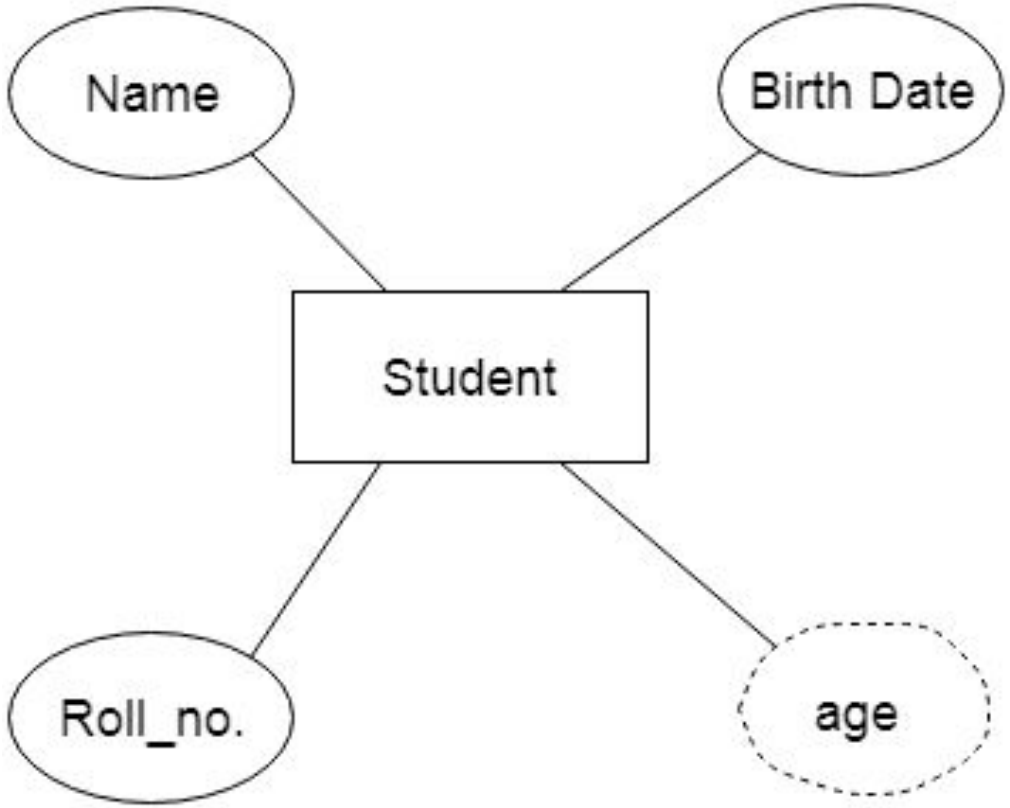
For example, a student can have more than one phone number.



d. Derived Attribute

An attribute that can be derived from other attribute is known as a derived attribute. It can be represented by a dashed ellipse.

For example, A person's age changes over time and can be derived from another attribute like Date of birth.



Types of relationship are as follows:

One-to-One Relationship

- When only one instance of an entity is associated with the relationship, then it is known as one to one relationship.
- **For example,** A female can marry one male, and a male can marry one female.



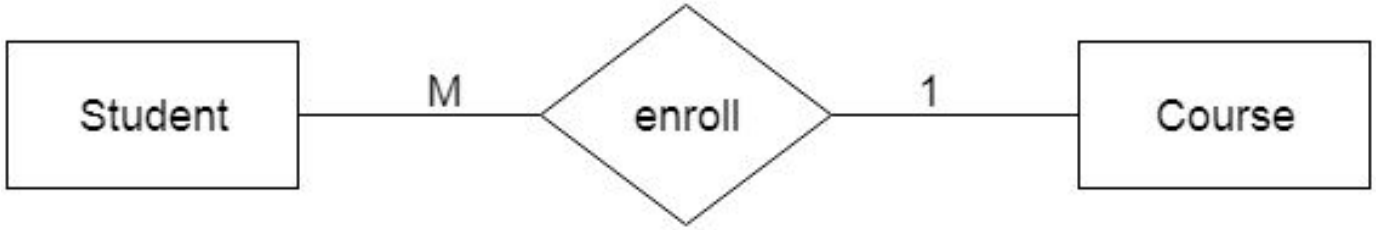
One-to-many relationship

- When only one instance of the entity on the left, and more than one instance of an entity on the right associates with the relationship then this is known as a one-to-many relationship.
- **For example,** Scientist can invent many inventions, but the invention is done by the only specific scientist.



Many-to-one relationship

- When more than one instance of the entity on the left, and only one instance of an entity on the right associates with the relationship then it is known as a many-to-one relationship.
- **For example,** Student enrolls for only one course, but a course can have many students.

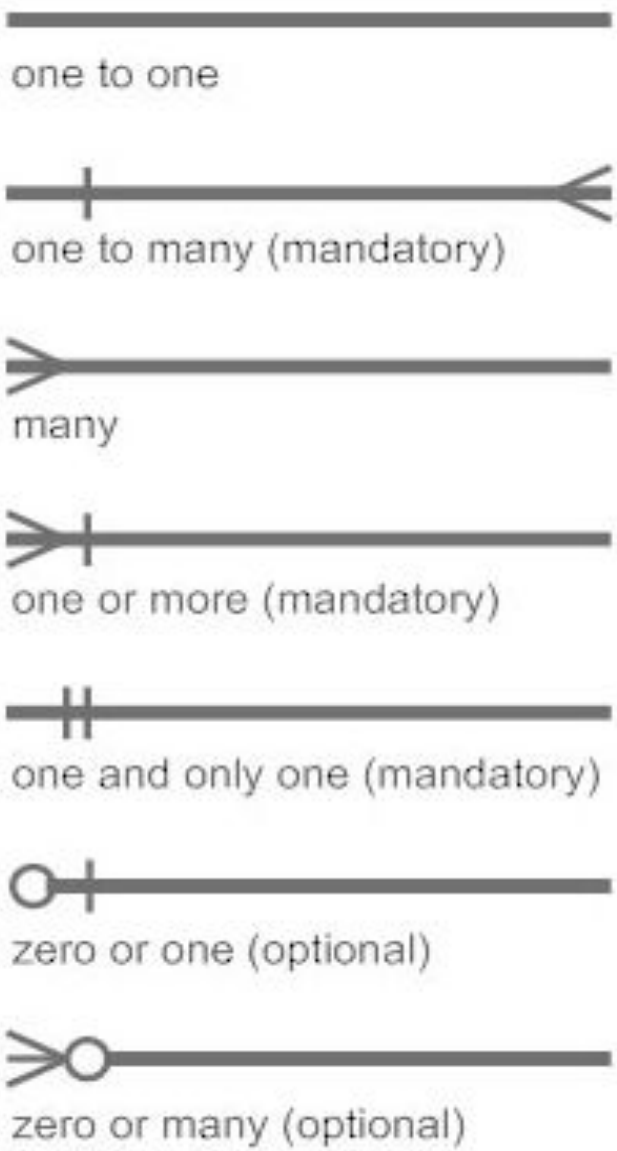


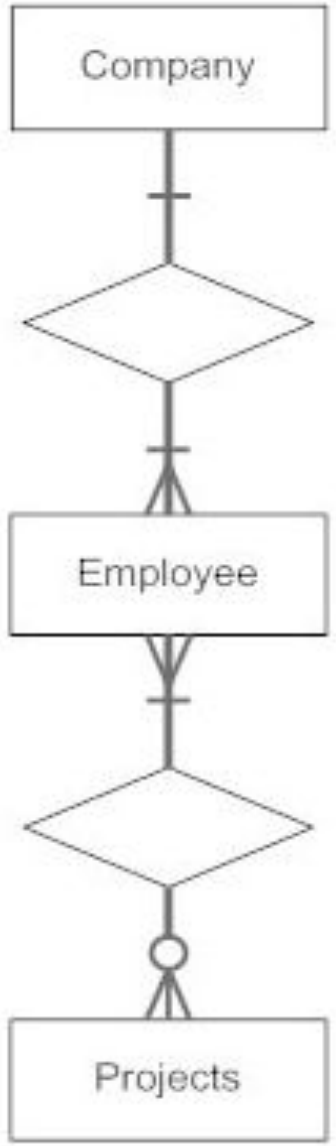
Many-to-many relationship

- When more than one instance of the entity on the left, and more than one instance of an entity on the right associates with the relationship then it is known as a many-to-many relationship.
- **For example,** Employee can assign by many projects and project can have many employees.



Notations


one to one
one to many (mandatory)
many
one or more (mandatory)
one and only one (mandatory)
zero or one (optional)
zero or many (optional)


Company
Employee
Projects

Cardinality in DBMS (Mapping Constraints)

Cardinality

- Cardinality means how the entities are arranged to each other or what is the relationship structure between entities in a relationship set.
- In a Database Management System, Cardinality represents a number that denotes how many times an entity is participating with another entity in a relationship set.
- The Cardinality of DBMS is a very important attribute in representing the structure of a Database.
- In a table, the number of rows or tuples represents the Cardinality.

Cardinality Ratio

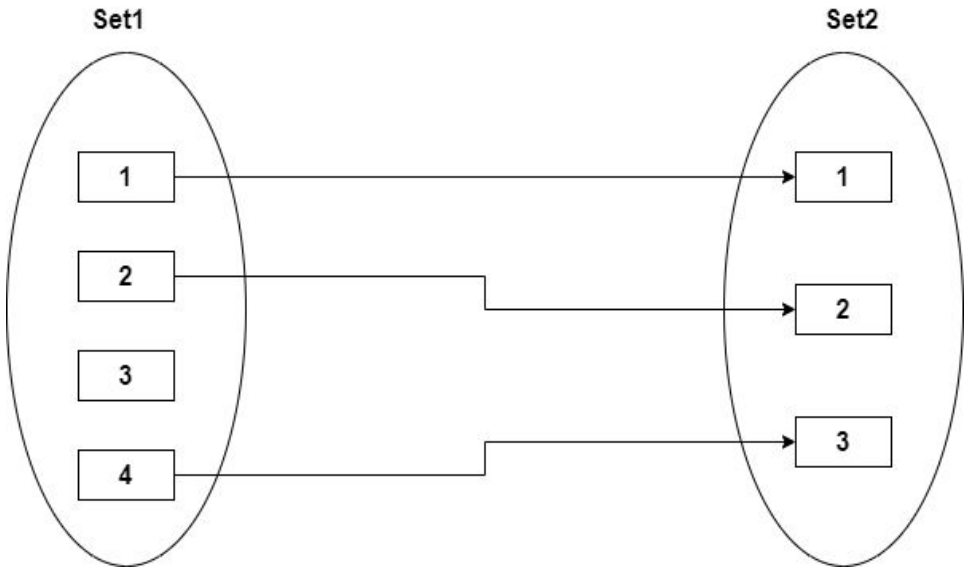
- Cardinality ratio is also called **Cardinality Mapping**, which represents the mapping of one entity set to another entity set in a relationship set.
- We generally take the example of a binary relationship set where two entities are mapped to each other.

There are four types of Cardinality Mapping in Database Management Systems:

- One to one
- Many to one
- One to many
- Many to many

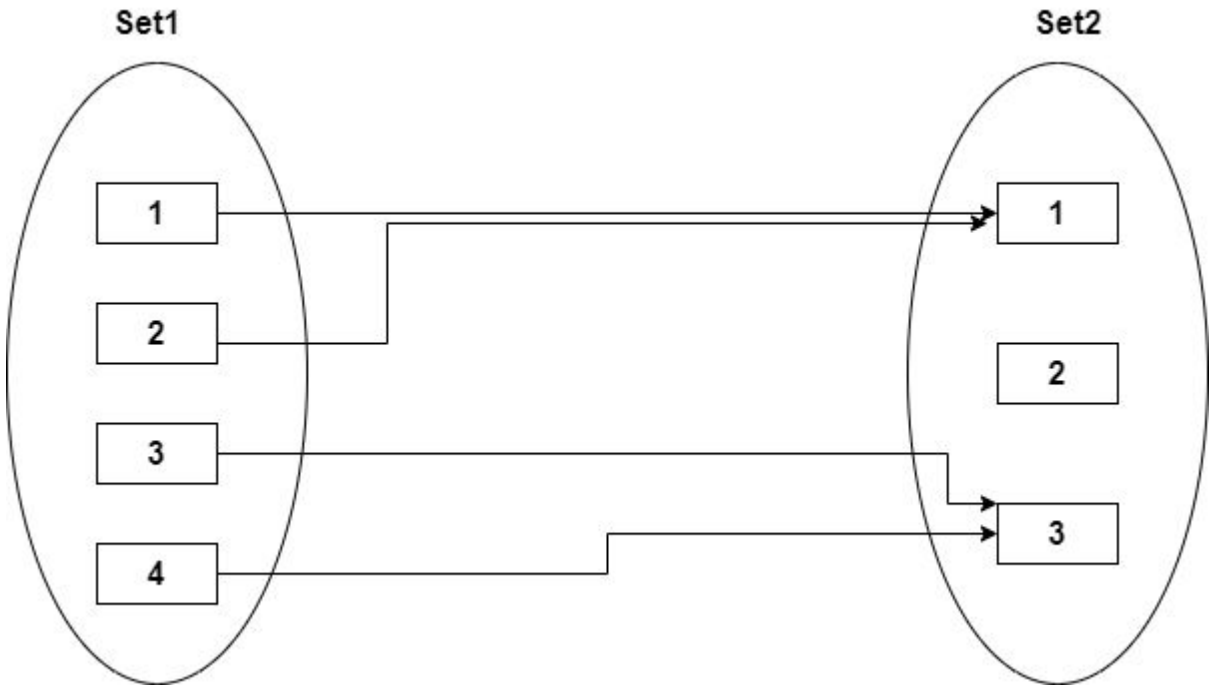
One to One

- One to one cardinality is represented by a **1:1** symbol. In this, there is at most one relationship from one entity to another entity. There are a lot of examples of one-to-one cardinality in real life databases.
- **For example,** one student can have only one student id, and one student id can belong to only one student. So, the relationship mapping between student and student id will be one to one cardinality mapping.



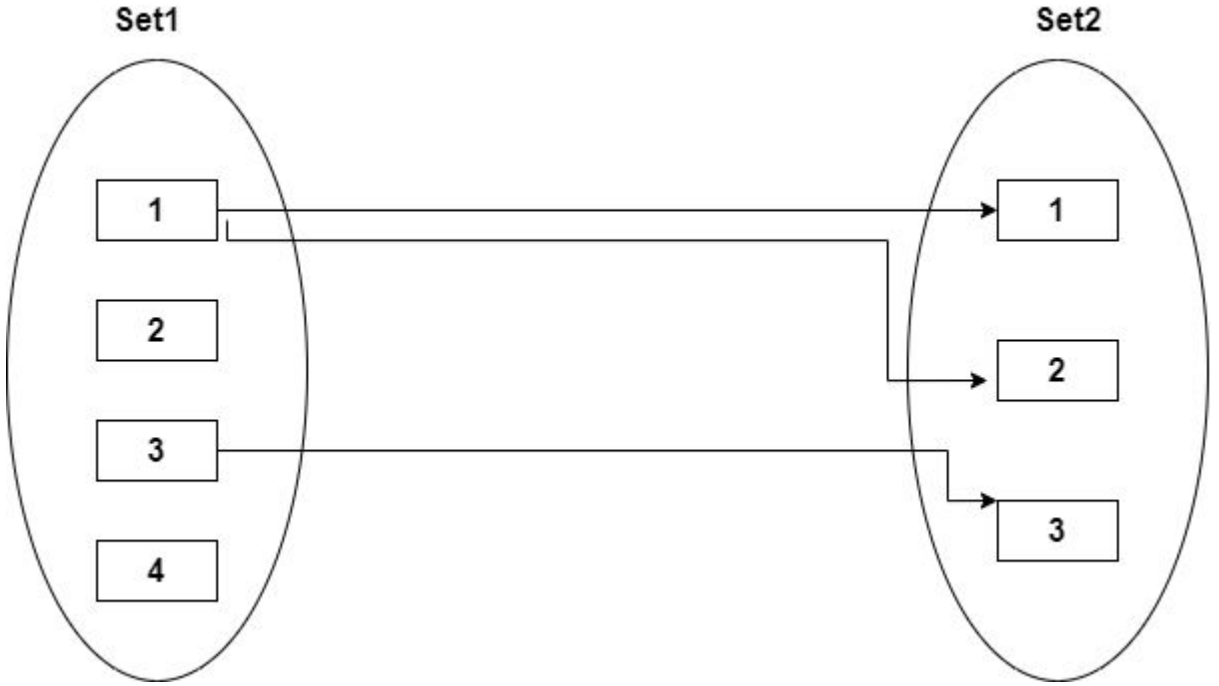
Many to One Cardinality:

- In many to one cardinality mapping, from set 1, there can be multiple sets that can make relationships with a single entity of set 2. Or we can also describe it as from set 2, and one entity can make a relationship with more than one entity of set 1.
- One to one Cardinality is the subset of Many to one Cardinality. It can be represented by **M:1**.
- **For example**, there are multiple patients in a hospital who are served by a single doctor, so the relationship between patients and doctors can be represented by Many to one Cardinality.



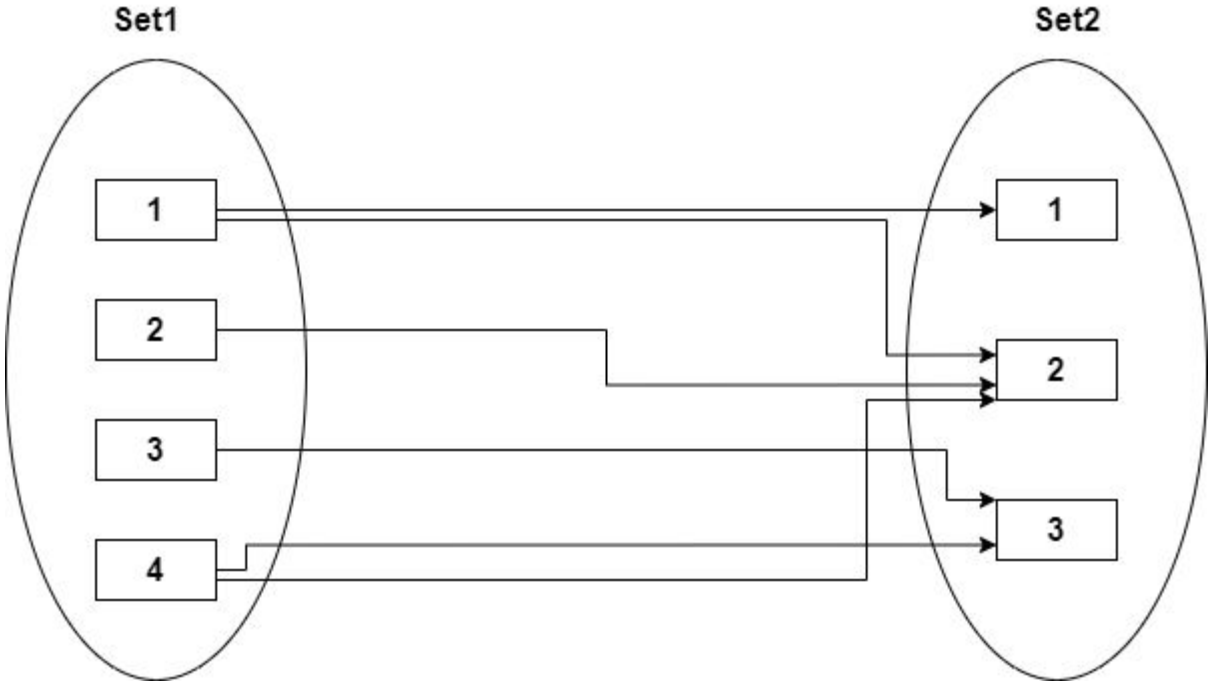
One to Many Cardinalities:

- In One-to-many cardinality mapping, from set 1, there can be a maximum single set that can make relationships with a single or more than one entity of set 2. Or we can also describe it as from set 2, more than one entity can make a relationship with only one entity of set 1.
- One to one cardinality is the subset of One-to-many Cardinality. It can be represented by **1: M**.
- **For Example,** in a hospital, there can be various compounders, so the relationship between the hospital and compounders can be mapped through One-to-many Cardinality.



Many to Many Cardinalities:

- In many, many cardinalities mapping, there can be one or more than one entity that can associate with one or more than one entity of set 2. In the same way from the end of set 2, one or more than one entity can make a relation with one or more than one entity of set 1.
- It is represented by **M: N** or **N: M**.
- One to one cardinality, One to many cardinalities, and Many to one cardinality is the subset of the many to many cardinalities.
- **For Example,** in a college, multiple students can work on a single project, and a single student can also work on multiple projects. So, the relationship between the project and the student can be represented by many to many cardinalities.

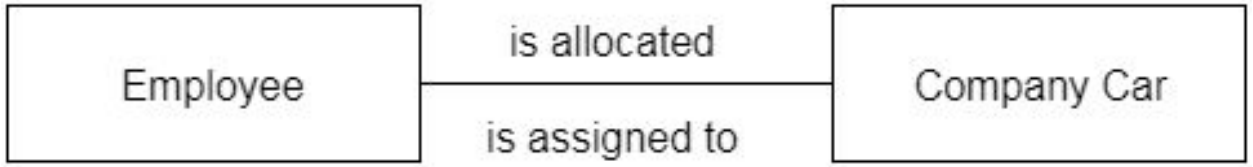


Relationship of higher degree

- The degree of relationship can be defined as the number of occurrences in one entity that is associated with the number of occurrences in another entity.
- There is the three degree of relationship:
  - One-to-one (1:1)
  - One-to-many (1:M)
  - Many-to-many (M:N)

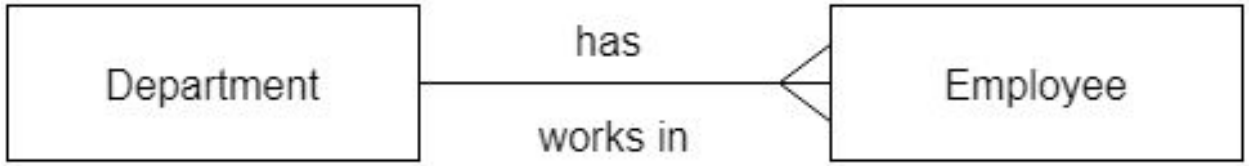
One-to-one

- In a one-to-one relationship, one occurrence of an entity relates to only one occurrence in another entity.
- A one-to-one relationship rarely exists in practice.
- **For example:** if an employee is allocated a company car then that car can only be driven by that employee.
- Therefore, employee and company car have a one-to-one relationship.



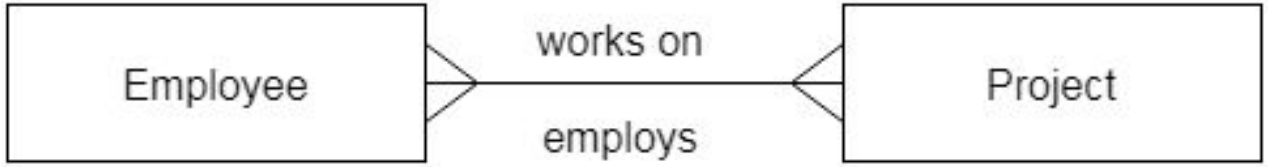
One-to-many

- In a one-to-many relationship, one occurrence in an entity relates to many occurrences in another entity.
- **For example:** An employee works in one department, but a department has many employees.
- Therefore, department and employee have a one-to-many relationship.

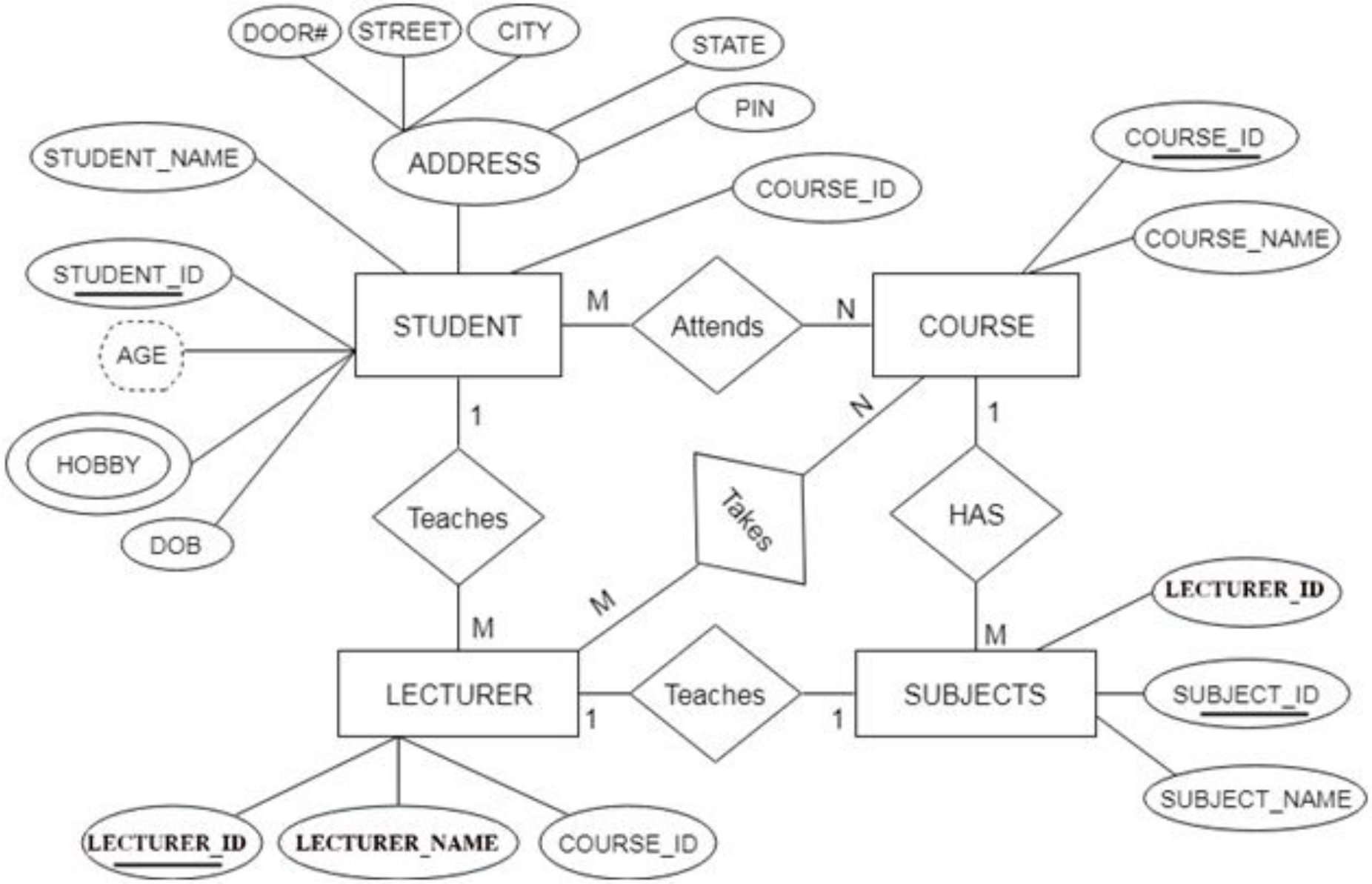


Many-to-many

- In a many-to-many relationship, many occurrences in an entity relate to many occurrences in another entity.
- Same as a one-to-one relationship, the many-to-many relationship rarely exists in practice.
- **For example:** At the same time, an employee can work on several projects, and a project has a team of many employees.
- Therefore, employee and project have a many-to-many relationship.



Reduction of ER diagram to Table



There are some important points in converting E-R diagram into Table:

Entity type becomes a table.

- In the given ER diagram, LECTURE, STUDENT, SUBJECT and COURSE forms individual tables.

All single-valued attribute becomes a column for the table.

- In the STUDENT entity, STUDENT_NAME and STUDENT_ID form the column of STUDENT table.
- Similarly, COURSE_NAME and COURSE_ID form the column of COURSE table and so on.

A key attribute of the entity type represented by the primary key.

- In the given ER diagram, COURSE_ID, STUDENT_ID, SUBJECT_ID, and LECTURE_ID are the key attribute of the entity.

The multivalued attribute is represented by a separate table.

- In the student table, a hobby is a multivalued attribute. So it is not possible to represent multiple values in a single column of STUDENT table.
- Hence we create a table STUD_HOBBY with column name STUDENT_ID and HOBBY. Using both the column, we create a composite key.

Composite attribute represented by components.

- In the given ER diagram, student address is a composite attribute. It contains CITY, PIN, DOOR#, STREET, and STATE. In the STUDENT table, these attributes can merge as an individual column.

Derived attributes are not considered in the table.

- In the STUDENT table, Age is the derived attribute. It can be calculated at any point of time by calculating the difference between current date and Date of Birth.

Table structure:

